Score-based diffusion models

The mathematics behind modern generative models

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1. The problem

Goal: generate new samples from an unknown data distribution $p_0(x)$.

We progressively corrupt observed samples with noise until reaching a known prior π, then learn how to reverse the process.

2. Diffusions as generative models

The forward process is an ItÃt' SDE:

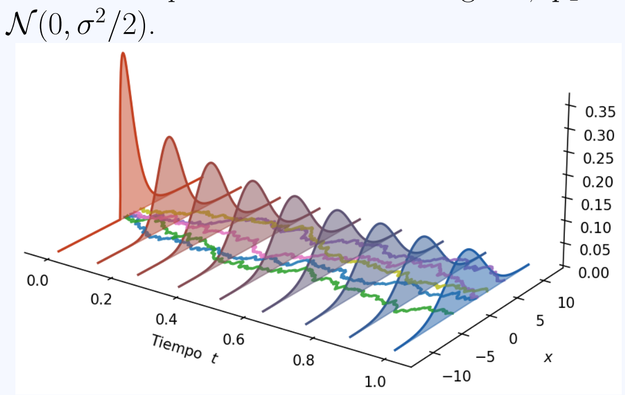
$$\mathrm{d}X_t = f(X_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}B_t, \qquad X_0 \sim p_0.$$

For a suitable drift and noise schedule, p_T approaches a tractable Gaussian prior. Anderson's time-reversal result adds the score to the reverse drift.

3. Forward process: Ornstein–Uhlenbeck

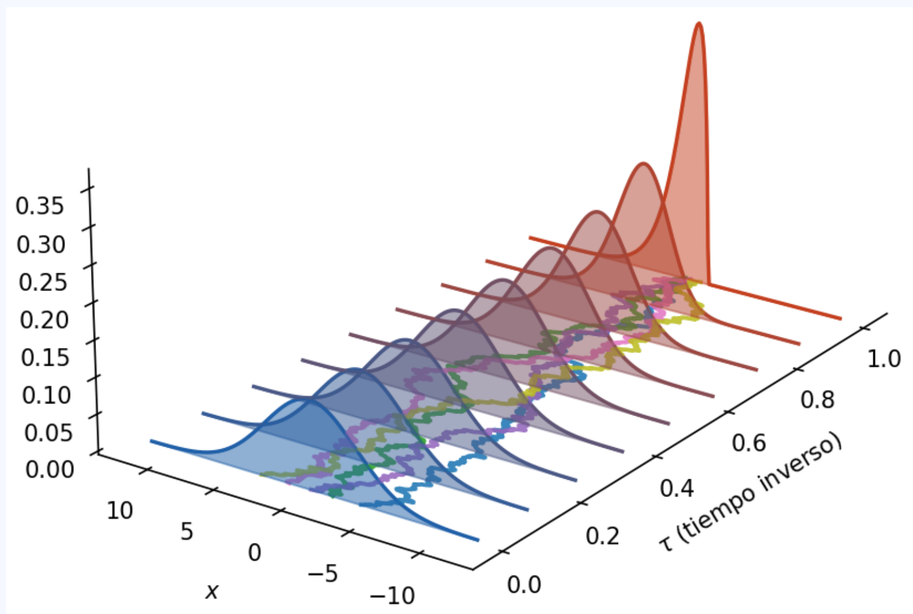
$$\mathrm{d}X_t = -X_t\,\mathrm{d}t + \sigma\,\mathrm{d}B_t.$$

The stable drift contracts the process while Brownian noise spreads mass. For large T, $p_T \approx \mathcal{N}(0, \sigma^2/2)$.



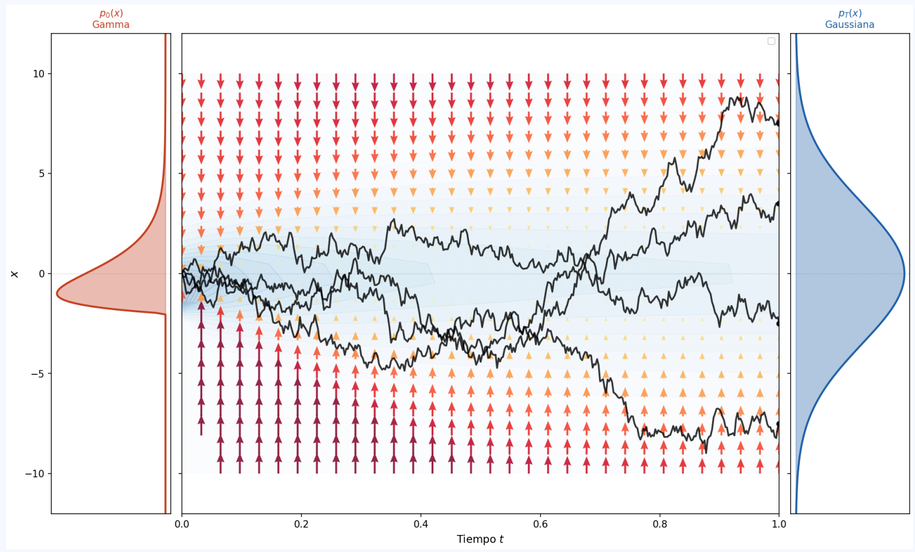
4. Anderson (1982): reverse time

$$\mathrm{d}\hat{X}_\tau = \left[\hat{X}_\tau + \sigma^2\,\nabla_x \log p_{T-\tau}\right]\,\mathrm{d}\tau + \sigma\,\mathrm{d}\bar{B}_\tau.$$



5. The score

The unknown field $\nabla_x \log p_t(x)$ is the score. It points toward higher-probability regions and supplies the correction that guides reverse diffusion.



Learning target: $s_\theta(x, t) \approx \nabla_x \log p_t(x)$.

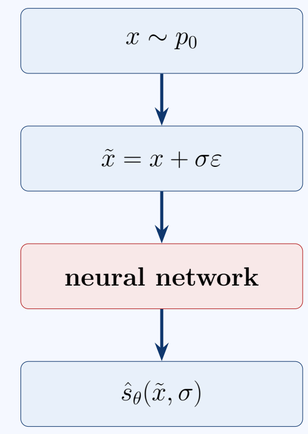
6. Learning the score

The conditional score of a Gaussian corruption kernel is known:

$$\nabla_{\tilde{x}} \log q_\sigma(\tilde{x} \mid x) = -\frac{\tilde{x} - x}{\sigma^2}.$$

Denoising score matching trains a neural network against this target across noise levels. The network learns a vector field, not a normalized density.

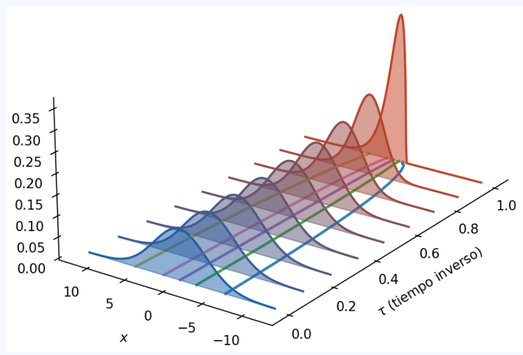
7. What does the network learn?



8. Probability-flow ODE

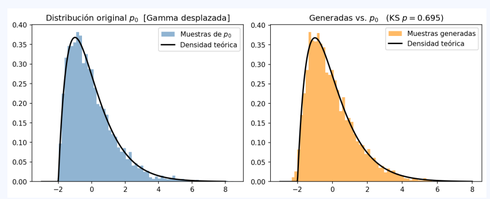
The reverse dynamics also admit a deterministic ODE with the same marginals:

$$\mathrm{d}\hat{X}_\tau = \left[\hat{X}_\tau + \tfrac{\sigma^2}{2}\nabla_x \log p_{T-\tau}\right]\,\mathrm{d}\tau.$$



9. The pipeline works

Sample from the Gaussian prior, integrate the reverse SDE or probability-flow ODE, and obtain an approximation to p_0.



KS test: $p = 0.695$; the null is not rejected.

Geometry + probability + deep learning.

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